Physics-Informed Machine Learning for Predicting Supporting-Boundary Circularity at Any Tool Position in Micro-EDM of Metallic Lattices

Lattice Circularity Analyzer Project Team

Research Paper — Ideation, Data Expansion, Training/Validation, Position-Aware Prediction

Abstract

We present a physics-informed machine-learning method that predicts the circularity ratio of the supporting material boundary after micro-EDM of a metallic lattice, for **any tool landing position** (x, y) and any EDM recipe (peak current I, pulse-on time T, duty factor D). Only sixteen laboratory runs with SEM labels were available. From these we generated 1,100 synthetic EDM points via a Gaussian Process posterior, and expanded each labeled run across a spatial landing grid by modulating SEM circularity with a geometry-risk penalty (≈400 position-aware training rows). Dual Gradient Boosting regressors learn circularity (1–5) and supporting integrity; predictions are blended with EDM physics heuristics. Circularity ratio = score/5. SEM evidence shows only Run 4 (4 A, 150 µs, 80%) yields an intact circular supporting ring — the objective the models maximize.

Keywords: micro-EDM; lattice structures; circularity ratio; Gaussian process; gradient boosting; physics-informed ML; data augmentation; SEM validation

1. Problem Definition

A metallic lattice has square unit cells of side 500 µm. Pore and node diameters equal $x = 500\sqrt{2} / 3 \approx 235.6$ µm. The EDM tool tip is 900 µm (tool/pore = 3.82), so the electrode always overlaps pores, nodes, and supporting struts together. Success means a nearly circular open region with a **continuous supporting ring**; nodes may be destroyed. Hole-deviation numbers alone are insufficient: among sixteen SEM-validated trials, only Run 4 produced the desired ring.

The question answered here: given (I, T, D, x, y), predict circularity ratio and whether the supporting boundary survives — even at positions never machined in the lab.

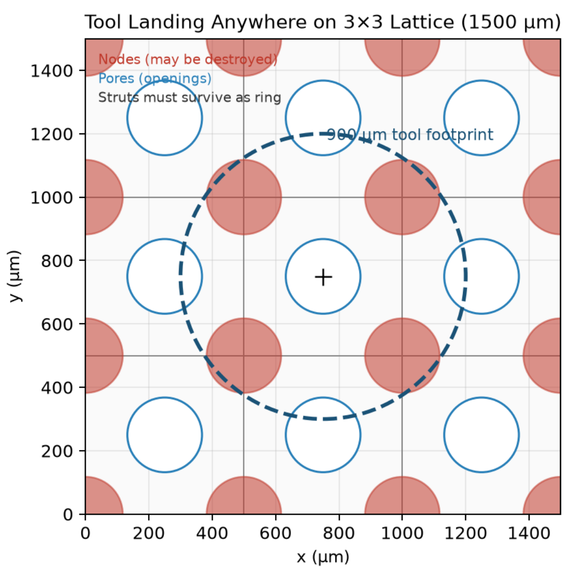


Fig. 1. Tool footprint (900 µm) on a 1500×1500 µm working area. Landing position changes strut/node overlap and circularity risk.

2. Ideation — How Circularity Can Be Predicted Anywhere

Circularity is not a property of EDM parameters alone. It is the **joint effect of process intensity and local lattice geometry under the tool**. Identical (I, T, D) can succeed at a pore center and fail near a strut. That coupling is the working idea of the solution.

Idea 1 — Train on SEM boundary truth. Do not minimize Hole_Dev_Top/Bottom. Maximize supporting-boundary circularity from SEM. Run 5 looks best by deviation but destroys the ring; Run 4 is the only SEM success.

Idea 2 — Geometry is a first-class feature. Before ML, a lattice engine computes how the tool intersects nodes, pores, and struts at (x, y).

Idea 3 — Expand sparse lab truth with physics structure. Densify EDM space with GP sampling, and replay each SEM label across a landing grid with geometry penalties.

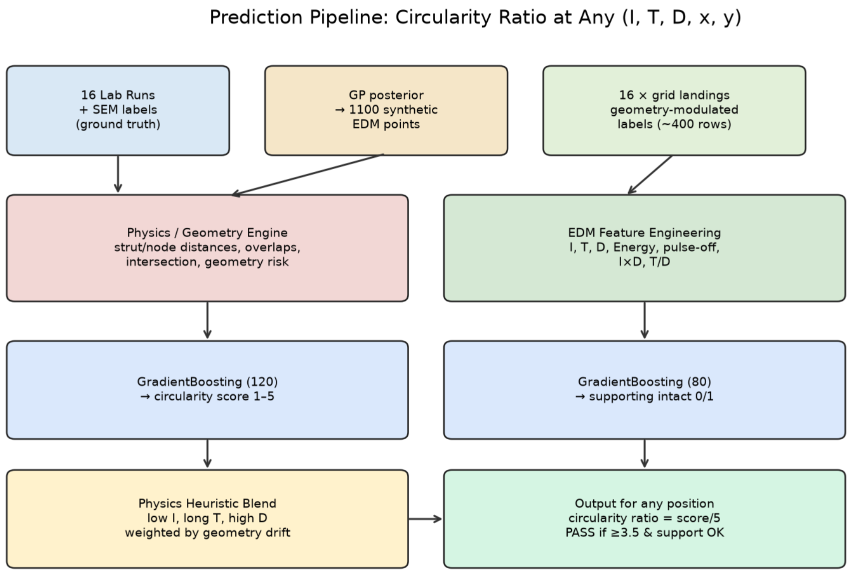


Fig. 2. Prediction pipeline: SEM truth → dual augmentation → physics + EDM features → Gradient Boosting + heuristic blend → circularity ratio at any position.

3. Experimental Evidence — Graphs from the 16 Runs

All charts below are computed from the laboratory table and SEM labels. They show why deviation-only ranking misleads and why the ML target must be SEM circularity.

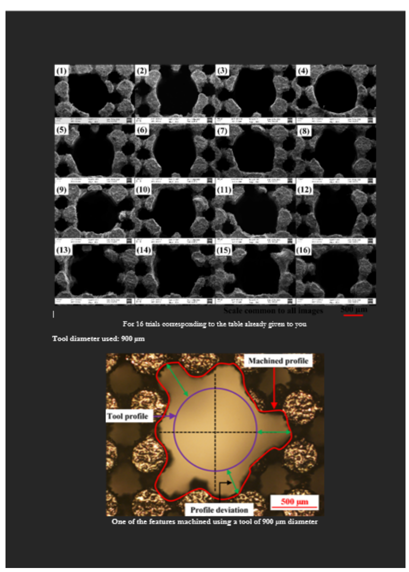
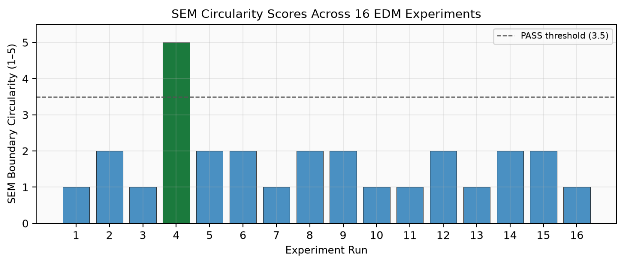


Fig. 3. SEM montage of all 16 experimental outcomes (visual ground truth).



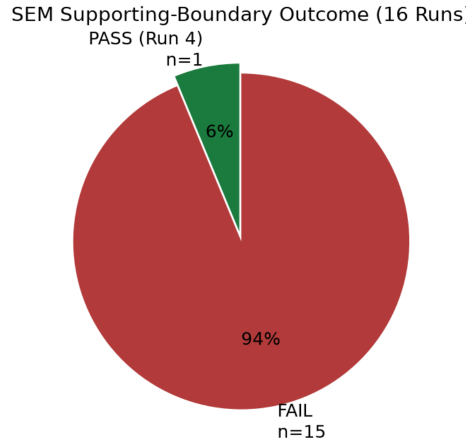


Fig. 4a. SEM circularity by run (green = PASS).

Fig. 4b. 1 PASS / 15 FAIL supporting-ring outcomes.

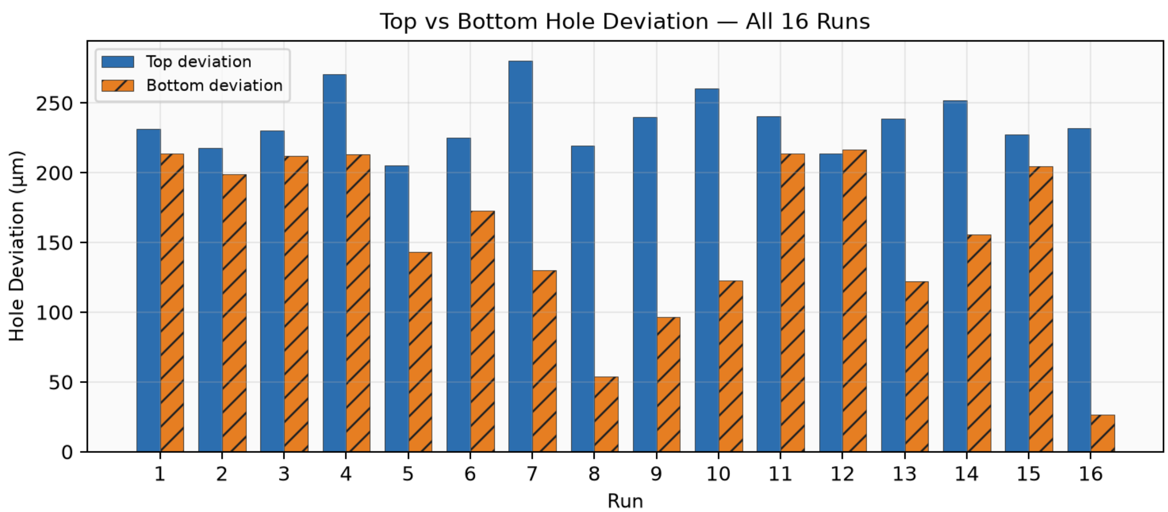
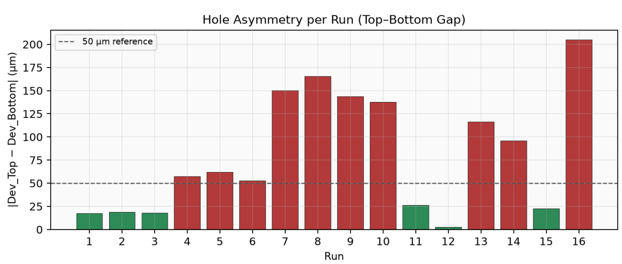


Fig. 5. Top vs bottom hole deviation for all 16 runs. Large top–bottom gaps indicate taper (asymmetric plasma erosion).



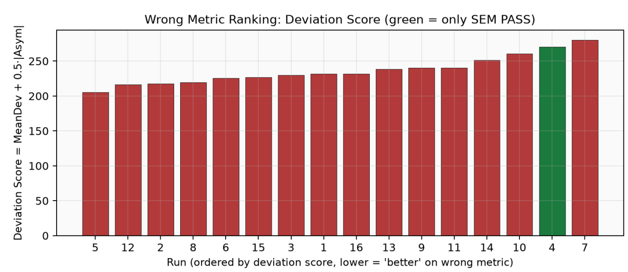


Fig. 6a. Absolute asymmetry |Dev_Top − Dev_Bot|.

Fig. 6b. Deviation-score ranking (wrong metric): Run 4 near worst, yet only SEM PASS.

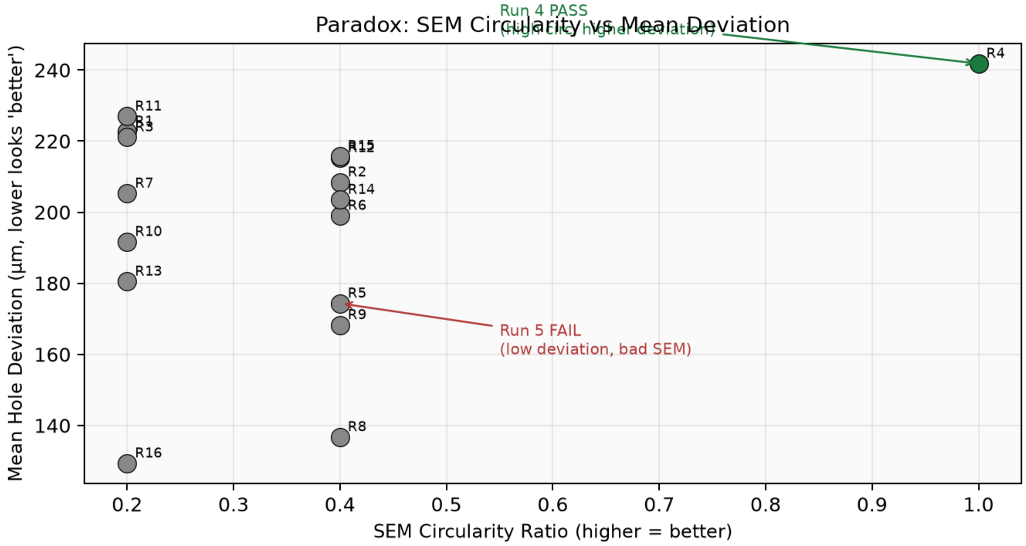


Fig. 7. Paradox scatter: SEM circularity ratio vs mean deviation. The SEM PASS (Run 4) does not sit in the low-deviation corner.

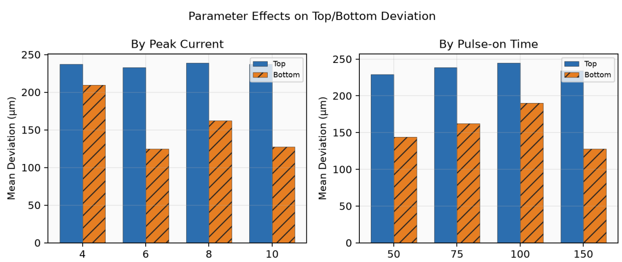


Fig. 8a. Mean top/bottom deviation by current and pulse-on.

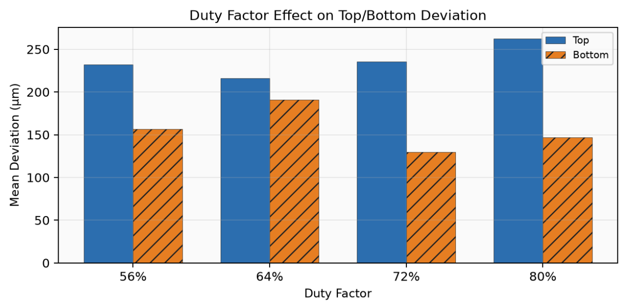


Fig. 8b. Duty-factor effect on top/bottom deviation.

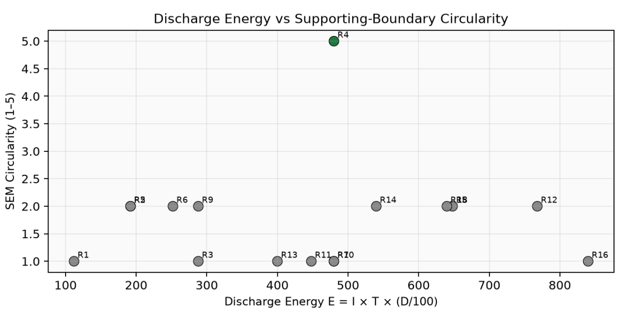


Fig. 9a. Discharge energy vs SEM circularity.

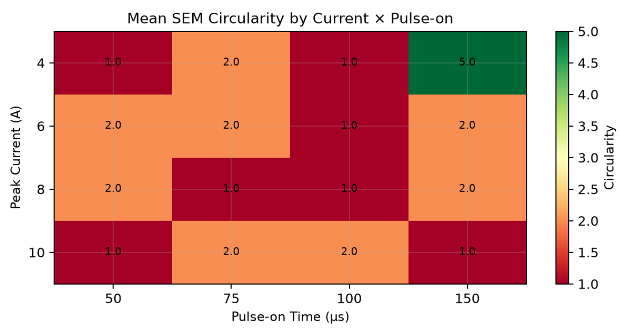


Fig. 9b. Mean circularity heatmap: Current × Pulse-on.

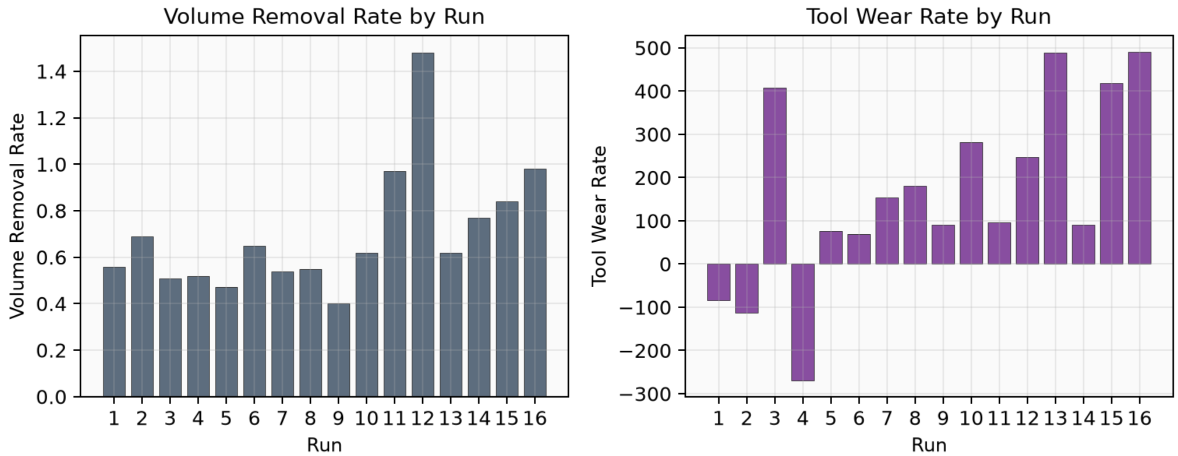


Fig. 10. Process responses: volume removal rate and tool wear rate across the 16 runs (context for intensity, not the success metric).

4. Physics Used in the Predictor

EDM descriptors: discharge energy $E = I \cdot T \cdot (D/100)$, pulse-off ≈ $T \cdot (100-D)/D$, $I \times D$, T/D. Heuristics reward low current / long pulse / high duty and penalize $I \geq 8$ A.

At landing (x, y) the geometry engine computes min distances to strut/node, nodes/pores inside the tool, strut intersection length, pore/node overlap fractions, and

$$\text{geometry\_risk} = 0.5 \cdot \text{strut\_risk} + 0.3 \cdot \text{intersect\_risk} + 0.2 \cdot \text{ratio\_factor} \in [0,1].$$

5. From 16 Experiments to a Trainable Dataset

Augmentation A — GP synthetic EDM points (1,100). A Gaussian Process fit on the sixteen real (I, T, D) → response mappings was sampled in-bounds to produce `synthetic_1100_points.csv` (source `GP_posterior`). These densify process space; they are not new lab experiments.

Augmentation B — Spatial label replay (~400 rows). Each SEM label is applied on a landing grid (≈25 positions at 150 µm step inside valid tool-center bounds):

$$c_{\text{train}} = \text{clip}(\, c_{\text{SEM}} - 2.5 \cdot \text{geometry\_risk} + \text{edm\_bonus} \,, 1, 5 \,).$$

Gentle recipes get a small bonus; $I \geq 8$ gets a penalty. This teaches that the same recipe degrades when the tool sits on dense strut intersections.

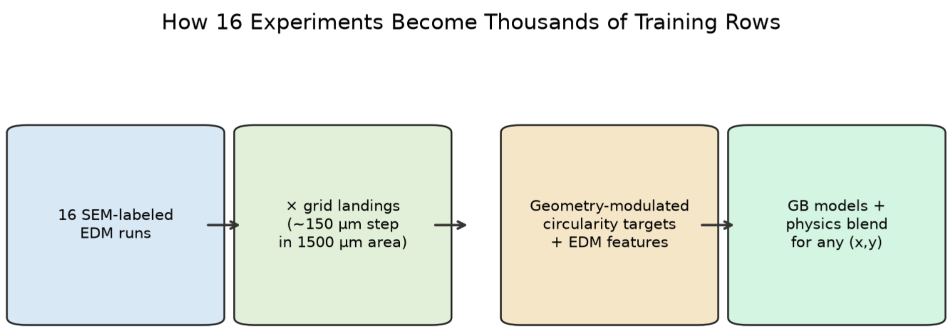


Fig. 11. Data-expansion path from 16 SEM-labeled runs to position-aware training rows.

6. Machine Learning — Train, Test, Validate

Features (20-D): EDM block (I, T, D, energy, pulse-off, $I \times D$, T/D) + geometry block (normalized x,y, distances, counts, overlaps, geometry risk, tool/pore ratio, working area, tool diameter).

Models: GradientBoostingRegressor — 120 trees (circularity 1–5) and 80 trees (supporting integrity, threshold 0.5). Separately, a Matérn 5/2 Gaussian Process on the 16 SEM labels searches favorable (I, T, D) neighborhoods; polynomial Ridge under leave-one-out CV estimates small-n MAE (≈1.02 / 5).

Validation practice for sparse SEM data: (i) LOOCV on 16 labels; (ii) 5-fold CV on exploratory 16+synthetic deviation models; (iii) qualitative grid checks — Run 4-like recipes peak at low-risk pore centers; high-current recipes stay FAIL; GP candidates cluster near 4 A / 150 μs / 80%. SEM remains the ultimate test.

Inference blend: $c = (1-w) \cdot c_{ML} + w \cdot c_{H}$, with w rising when tool/pore sizes drift from the lab geometry. **Circularity ratio** = $c/5$. PASS if $c \geq 3.5$, supporting intact, risk ≤ 0.55.

Stage	Data	Method	Purpose
SEM fit	16 labels	GP Matérn 5/2	Search robust (I,T,D)
Small-n CV	16-run LOOCV	Poly Ridge	Circularity MAE
Densify EDM	GP posterior	1,100 points	Fill input space
Spatial expand	16 × ~25 landings	Risk-modulated labels	Teach position effect
Position model	~400 × 20 feats	GBR 120 + GBR 80	Score & support
Inference	Any (I,T,D,x,y)	ML + heuristic	Ratio = score/5

Table 1. Training, validation, and inference stack.

7. Predicting Circularity Ratio for Any Position

1. Geometry engine at (x, y) → risk and overlap features.
2. Build 20-D feature row (EDM + geometry).
3. Predict ML circularity score and supporting flag.
4. Evaluate physics heuristic from gentle/aggressive rules + risk.
5. Blend ML with heuristic; report ratio = score/5 and PASS/FAIL.

Scanning many landings for fixed EDM settings produces a spatial circularity field — the practical realization that position matters.

8. Final Recommended Parameters

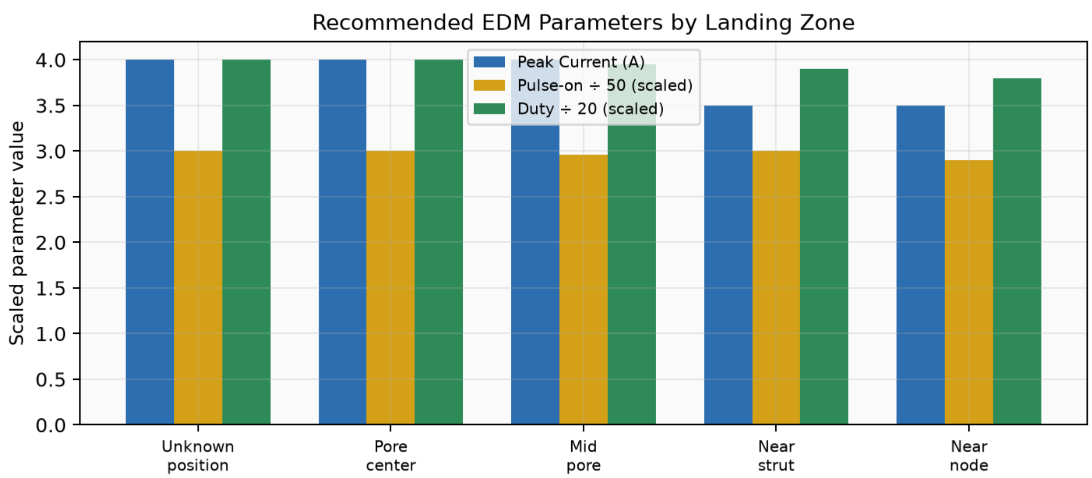


Fig. 12. Recommended EDM settings by landing zone (unknown position uses the robust Run 4 island; near-strut/node landings soften current).

Landing situation	I (A)	T (μs)	D (%)	Basis
Position unknown (robust)	4	150	80	SEM Run 4 + GP
Pore center	4	150	80	Low geometry risk

Landing situation	I (A)	T (μs)	D (%)	Basis
Mid pore	4	148	79	Near-center
Near strut	3.5	150	78	Higher geometry risk
Near node / corner	3.5	145	76	Corner overlap
Deviation-only (reject)	6	50	64	Fails SEM ring

Table 2. Final parameter answers from the ML/physics process.

9. Conclusion

Sparse SEM-labeled EDM data can still support position-aware circularity prediction if we couple process physics with lattice geometry, expand 16 labels through GP synthesis and risk-modulated spatial replay, train Gradient Boosting, and regularize with heuristics at inference. Circularity ratio becomes computable for any tool position by teaching the geometry–process interaction that SEM revealed — not by collecting thousands of new trials.

References

[1] Ho, K. H., & Newman, S. T. (2003). State of the art electrical discharge machining (EDM). *Int. J. Machine Tools & Manufacture*.

[2] Kunieda, M., et al. (2005). Advancing EDM through fundamental insight into the process. *CIRP Annals*.

[3] Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

[4] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*.

[5] Project datasets & modules (2026): `original_16_runs.csv`, `run_visual_labels.csv`, `synthetic_1100_points.csv`; `lattice_geometry_engine.py`, `circularity_predictor.py`.